

AI Persona - Evaluation Report

Yashraj Kupekar | AI Engineer Intern Assignment

1. Voice Quality Measurement

Voice agent built with Vapi (Deepgram nova-3 STT, ElevenLabs TTS, GPT-4o). Measured across 10+ live test calls on a provisioned US phone number.

STT Model	Deepgram nova-3 (54% lower WER vs nova-2)
Locale	en-IN (Indian English phonetics)
First Response Latency	< 2s (Vapi-managed pipeline)
Interruption Handling	stopSpeakingPlan: 2-word threshold, 0.3s voice gate
Turn-Taking	smartEndpointingEnabled + 0.6s silence wait
Noise Handling	backgroundDenoisingEnabled: true

Key optimizations: keyterm prompting for multi-word phrases ("Yashraj Kupekar", "Medicine Voice AI"), keyword boosting (weight 3-5) for domain terms, chunkPlan with punctuation boundaries for low-latency TTS streaming.

2. Chat Groundedness Measurement

Evaluated with RAGAS (4 metrics, 8 test questions) + custom eval categories (29 total tests, 100% pass rate):

RAGAS Faithfulness	0.784 (response grounded in retrieved context)
RAGAS Answer Relevancy	0.856 (response addresses the question)
RAGAS Context Precision	1.000 (retrieved context is relevant)
RAGAS Context Recall	0.792 (relevant info is retrieved)
Hallucination Detection	3/3 passed (LLM-as-judge)
Groundedness (tool usage)	5/5 passed (agent always calls tools first)
Tool Routing	4/4 passed (correct tool selected)
Refusal & Prompt Injection	4/4 passed (off-topic/injection redirected)
Retrieval Quality (Pinecone)	4/4 passed (term-matching validation)
Multi-turn Coherence	3/3 passed (3-turn conversation, LLM-judged)
Avg Chat TTFT	~5s Max: ~8s (includes Pinecone + GPT-4o)

3. Three Failure Modes Found & Fixed

Failure 1: Agent hallucinated project features (Faithfulness)

The agent claimed Medicine Voice AI "generates receipts" -- not in the knowledge base. Fix: Strict grounding instruction in the system prompt: "Never add details or claims not explicitly in tool output." RAGAS faithfulness improved from 0.4 to 0.784.

Failure 2: Agent engaged with off-topic code requests (Refusal)

When asked "Write a Python script to sort a list", the agent wrote the code. Fix: Explicit refusal rule for code/task requests unrelated to Yashraj. Refusal eval improved from 67% to 100%.

Failure 3: Voice STT misrecognized Indian English (STT Accuracy)

"Who is Yashraj" transcribed as "Rodriguez arrives". Root cause: Deepgram defaulting to en-US. Fix: en-IN locale, nova-2 -> nova-3, keyterm prompting for expected phrases. Transcription accuracy significantly improved on subsequent test calls.

4. What I'd Improve with 2 More Weeks

- Async agent pipeline: Convert sync OpenAI calls to async. Add request queuing and rate limiting for production traffic.
- Semantic caching: Cache frequent queries with embedding similarity. Would cut TTFT from ~5s to <500ms for repeated patterns.
- Automated voice eval pipeline: Generate test audio with TTS, send to Vapi via API, measure e2e latency and transcription accuracy programmatically.
- Observability: LangSmith/Opik tracing for every agent run -- tool latency breakdown, token usage, cost monitoring. Alert on faithfulness drops.
- Multi-language: Hindi voice support via Sarvam AI or Deepgram multilingual. Handle Hinglish code-switching naturally.